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Properties and Metabolic Diversity of Microbial Communities in Soils Treated with Steam Sterilization Compared with Methyl Bromide and Chloropicrin Fumigations

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The effects of steam sterilization (SS) on soil microbial properties including metabolic diversity of the microbial communities were examined in a greenhouse compared with those of two fumigants, methyl bromide (MB) and chloropicrin (CP). The numbers of fungi decreased in all the treatments. Nitrifiers, both ammonium-oxidizing bacteria and nitrite-oxidizing bacteria, were severely affected by the SS and CP treatments, resulting in their virtual disappearance. The decrease in the levels of microbial biomass C and N after the treatments suggested that the SS and CP treatments eradicated the microorganisms more effectively than the MB treatment and that the influence of the former persisted until the end of the experiment, 4 months after the treatments. Accumulation of $\text{NH}_4\text{-N}$ was observed after the SS and CP treatments mainly due to the partial decomposition of the dead microorganisms and the marked decrease in the number of ammonium-oxidizing bacteria. The richness and average well color development (AWCD) values in the microbial communities after the SS treatment were evaluated by the carbon substrate utilization method using Biolog ECO MicroPlates. The values decreased markedly immediately after the treatment but showed a rapid recovery, while those after the CP treatment continued to decrease until the transplanting of tomato seedlings. The effect of the MB treatment on the soil microbial communities was less pronounced. The growth of the tomato plants was promoted by the SS and CP treatments due to the increase in the N supply at the initial stage of tomato growth.

Key Words: effects on soil microorganisms, fumigation, metabolic diversity, N mineralization, steam sterilization.

Methyl bromide has been widely used in Japan as one of the most effective soil fumigants because of its spectrum of activity against plant pathogens, nematodes, weeds, and insects and the fact that no resistance to it had developed (Kobara 1998). However, its use is going to be prohibited from 2005 in accordance with the Montreal Protocol on substances that deplete the ozone layer. The implementation of an alternative program for pest management to maintain current crop yields and quality is of high priority. Even though the expanded use of existing fumigants such as chloropicrin is considered to be a substitute, steam sterilization is a more promising method as a clean and less harmful alternative for soil disinfection in greenhouses.

It has been reported that steam sterilization induces an accumulation of $\text{NH}_4\text{-N}$ and soluble Mn in soils, which may be detrimental to the nutritional status of the plants (Shizuoka Natl. Agric. Exp. Stn. 1971; Sonneveld 1979;

Tanaka et al. 2001). High soil temperature exceeding 60 to 70°C during steam sterilization affects soil microorganisms including nitrifiers (Shizuoka Natl. Agric. Exp. Stn. 1971) as well as pathogens. However, little information is available about the effect of steam sterilization on the activity and diversity of soil microbial communities.

The purpose of this study was to investigate the magnitude and duration of the effects of steam sterilization on the soil microbial properties compared with methyl bromide and chloropicrin fumigations. In addition to conventional microbial properties such as numbers of bacteria, fungi, and nitrifiers, levels of microbial biomass C and N, and N mineralization pattern, metabolic diversity was analyzed by the Biolog plate method (Garland and Mills 1991), which has been used for the evaluation of the effect of some fumigants (Engelen et al. 1998; Toyota et al. 1999; Itoh et al. 2002) and solariza-

tion (Itoh et al. 2000) on the soil microbial communities.

MATERIALS AND METHODS

Experimental design. Twelve experimental plots ($1 \times 5 \text{ m}^2$) were set up in a greenhouse at Kochi University, Japan in 2000. A randomized complete block design was used with four types of soil treatments in triplicates. Before the treatments, bark compost was applied at the rate of 50 Mg ha^{-1} . The experimental plots were subjected to steam sterilization (SS) on September 19, 2000, and to treatments with methyl bromide (MB) on September 20, and chloropicrin (CP) on September 18, along with a plot without soil disinfestation as control (CT). The steam was introduced into the soil from the surface with a canvas hose using a mobile steam applicator (K-500N, T-AGRI Co., Ltd., Fukuoka, Japan) until the soil temperature at a depth of 30 cm rose above 60°C (Fig. 1). The soil temperature was kept at above 60°C for a few hours after the removal of the applicator. Methyl bromide (25 g m^{-2}) was applied to the soil by the cold-gas method (Van Berkum and Hoestra 1975). Chloropicrin was injected using an applicator at a depth of 15 cm and at a rate of 3 mL per hole at 30 cm intervals $\times$ 3 stripes (30 g m^{-2}). After each treatment, the plots were covered with polyethylene sheets until September 25. Tomato seedlings (*Lycopersicon esculentum* cv. House Momotaro) were transplanted in the plots at a 30 cm spacing on October 3. No chemical fertilizers, herbicides, and insecticides were applied throughout the growth period.

Soil sampling. Soil samples were collected from the plow layer at a depth of 0–15 cm, before the treatments on September 18 (Day –2), immediately after the SS treatment (Day 0), immediately and 3 d after the removal of the covering sheet (Days 5 and 8), at the time of transplanting (Day 12), and 1, 2, and 3 months after transplanting (Days 48, 82, and 118). Soil sampling at Day 0 was conducted only for the SS treatment and CT.

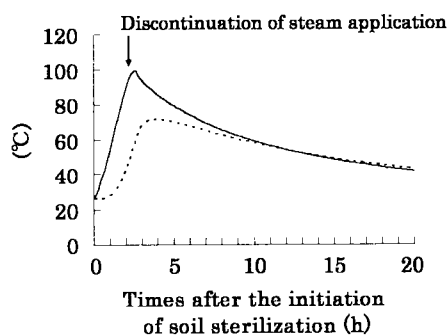


Fig. 1. Changes in soil temperature by the SS treatment. —, 10 cm depth; ---, 30 cm depth.

Soil samples were collected from two areas in each plot and were mixed well to obtain one composite sample. The composite samples were passed through a 4 mm mesh sieve and kept at 4°C for microbial analysis. The microbial analyses were conducted within 1 week after the sampling for the evaluation of microbial diversity and as early as possible for the other analyses. A portion of each composite sample was freeze-dried for chemical analysis. General physico-chemical properties of the soils at Day –2 were as follows: water content, 17%; pH (H_2O , 1 : 5), 6.26; EC (1 : 5), 0.0563 dS m^{-1} ; T-C, 14.1 g kg^{-1} ; T-N, 1.05 g kg^{-1} ; clay, 13.7%; silt, 16.0%; sand, 70.3%; CEC, $12.7 \text{ cmol}(+) \text{ kg}^{-1}$; and base saturation, 70.7%. These properties were rather homogeneous without a significant difference among the plots before the treatments.

Microbial properties. The viable counts of bacteria and fungi until Day 12 were performed by the plate count method after 5 d incubation at 28°C with 1 / 10 diluted nutrient broth medium for the bacteria and after 4 d incubation at 28°C with PDRB agar medium for the fungi (Lorch et al. 1995). The numbers of nitrifiers (ammonium-oxidizing and nitrite-oxidizing bacteria) were counted by the MPN method (Lorch et al. 1995). The levels of microbial biomass C and N were determined by the chloroform fumigation extraction method (Brookes et al. 1985; Vance et al. 1987).

Amounts of mineral N and N mineralization pattern. The amounts of $\text{NH}_4\text{-N}$ and $\text{NO}_3\text{-N}$ were determined by the steam distillation method (Keeney and Nelson 1982). Nitrogen mineralization of the soils collected at Days –2 and 12 was evaluated by the aerobic incubation method (Tanaka et al. 1998).

Metabolic diversity of microbial communities. To evaluate the diversity of the microbial communities, the substrate metabolism was investigated using the Biolog ECO MicroPlates (Biolog Inc., CA, USA). The plate has 96 wells containing 3 sets of 31 substrates and 1 control. The soil suspension was prepared by shaking with 0.1 M NaH_2PO_4 (pH 6) at a soil to solution ratio of 1 : 9 (w/v). Each well of the plates was inoculated with $100 \mu\text{L}$ of the soil suspension diluted to about 10^{-3} with 0.15 M NaCl. The inoculation was duplicated for one sample, and as a result, 6 replicates were prepared for one treatment. The plates were incubated at 28°C and hereafter their absorbance was measured at 590 nm using a microplate reader (Model 550, BIO-RAD, CA, USA) at regular time intervals. The absorbance values for the wells with C sources were blanked against the control well. Richness in metabolic diversity (Zak et al. 1994) was expressed as the average number of colored wells, where the absorbance after 3 d incubation exceeded 0.25. The values of the average well color development (AWCD; Garland 1996) were

calculated. The absorbance values after 3 d incubation were directly analyzed by principal component analysis using Excel Statistics 2000 for Windows program (SRI, Tokyo, Japan).

Plant growth. The growth of the tomato plants was surveyed 1 month after transplanting with 4 replications within one plot. The stem length and the number of developed leaves were measured. The second lower leaf below the flowering cluster was used for the determination of the SPAD value with a chlorophyll meter (SPAD-502 Chlorophyll meter, MINOLTA, Osaka, Japan), the leaf area measurement with an automatically area meter (AAM-7, Hayashidenkou, Tokyo, Japan), and the N content with a NC-analyzer (Sumigraph NC80, Sumika Chem. Anal. Service, Osaka, Japan).

Data analysis. All the data of the soils were expressed on an oven-dried basis. The effects of the soil treatments and the difference at each sampling time were compared by two-way analysis of variance, except for the samples collected at Day -2, which were expressed as an average of all the plots, and the samples at Day 0, which were collected only from SS and CT. The least significant difference (LSD) at the 5% level was calculated for the comparison of the treatments, sampling times, and any two samples in the scattergrams. The difference in the counts for bacteria and fungi and plant growth among the treatments were analyzed by Duncan's multiple range test at the 5% level.

RESULTS

General physico-chemical properties

No clear differences were observed in the general physico-chemical properties between the soils collected at Days -2 and 5, except for the water content and pH. The water content increased by 2% immediately after the SS treatment and then returned to the same level as that in the other treatments at Day 12. Soil pH in the SS treatment gradually increased until Day 12 and remained higher (about 0.2 to 0.3 units) than in the other treatments.

Microbial counts and biomass

The viable counts of fungi in the SS treatment decreased at Day 0 and recovered to the level of the CT at Day 12 while those in the MB and CP treatments tended to gradually decrease until Day 12 (Table 1). In contrast, the viable counts of bacteria did not show any considerable changes from Day -2 to Day 12, even though the counts in the SS, MB, and CP treatments were slightly higher than that in the CT. At Day 8, compared with Day -2, the level of microbial biomass C was reduced by 30% in the SS and 15% in the CP treatments, while the level of microbial biomass N was reduced by 74% in the SS, 58% in the MB, and 93% in the CP treatments (Fig. 2). Then, the levels of biomass C and N in the SS and CP treatments tended to be lower than those in the CT throughout the experiment, although there were

Table 1. Viable counts of bacteria and fungi.

Treatment	Day	Bacteria ($\log_{10}$ CFU g^{-1})					Fungi ($\log_{10}$ CFU g^{-1})				
		-2	0	5	8	12	-2	0	5	8	12
CT		7.76	7.72	6.95	6.71	7.16	5.53	5.26	5.33	4.48	4.63
SS		7.81	8.39	7.81	7.84	7.13	5.48	2.00	1.94	2.54	4.06
MB		8.23		7.10	7.54	7.45	5.57		4.34	3.17	2.83
CP		7.68		7.91	7.13	7.76	5.48		3.11	3.66	2.76

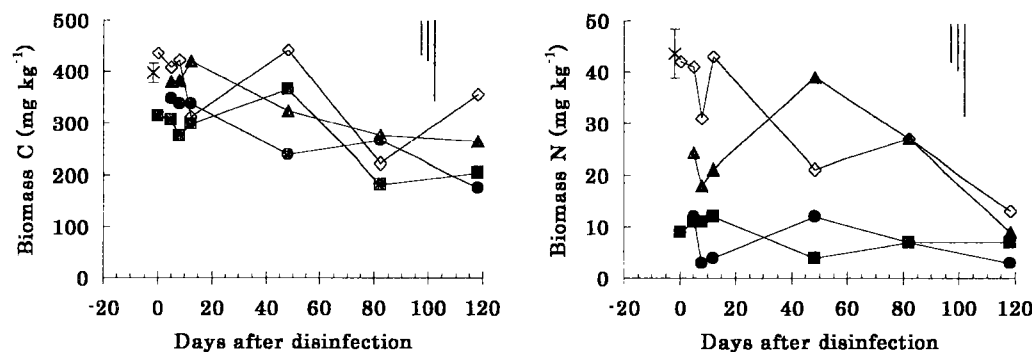


Fig. 2. Levels of microbial biomass C and N. $\times$, before disinfection; $\diamond$, CT; $\blacksquare$, SS; $\blacktriangle$, MB; $\bullet$, CP. The plot before disinfection indicates the average and standard deviation for all the plots. Vertical bars represent LSD ($p < 0.05$) for the comparison of the effect of soil treatments (left), sampling times (middle), and any pair of the results (right).

large fluctuations among the soil samples. The numbers of ammonium-oxidizing bacteria decreased after the three treatments. Especially those in the SS and CP treatments decreased substantially to about zero and remained lower than that in the CT until the end of the experiment (Fig. 3). On the other hand, the numbers of nitrite-oxidizing bacteria also decreased after the SS and CP treatments but recovered to the level of the CT faster than the numbers of ammonium-oxidizing bacteria.

Amounts of mineral N and N mineralization pattern

Considerable increase in the amounts of $\text{NH}_4\text{-N}$ occurred in the SS, CP, and MB treatments until Day 12 (Fig. 4). Thereafter, the levels of $\text{NH}_4\text{-N}$ slightly decreased in the MB treatment while they remained higher in the SS and CP treatments. Although the amounts of $\text{NO}_3\text{-N}$ varied widely at Day -2, the amounts decreased

after the SS and CP treatments while those in the MB treatment and CT remained almost unchanged.

The ammonification and nitrification patterns of the soils at Day -2, determined by the incubation method, were almost identical among the treatments (data not shown). The level of $\text{NH}_4\text{-N}$ was very low throughout the incubation, while the amount of $\text{NO}_3\text{-N}$ gradually increased along with incubation. In the soils collected at Day 12, except in the CT, the patterns were markedly different from those at Day -2 (Fig. 5). The levels of $\text{NH}_4\text{-N}$ of the soils in the SS and CP treatments markedly increased until 1 month of incubation while that in the MB treatment showed a relatively small increase. A sharp decrease in the level of $\text{NH}_4\text{-N}$ to that of the CT was observed after 84 d incubation for the SS and CP-treated soils and 21 d for the MB soil. There was a time lag of about 1 month before the initiation of nitrification in the soils subjected to the SS and CP treatments. Soil

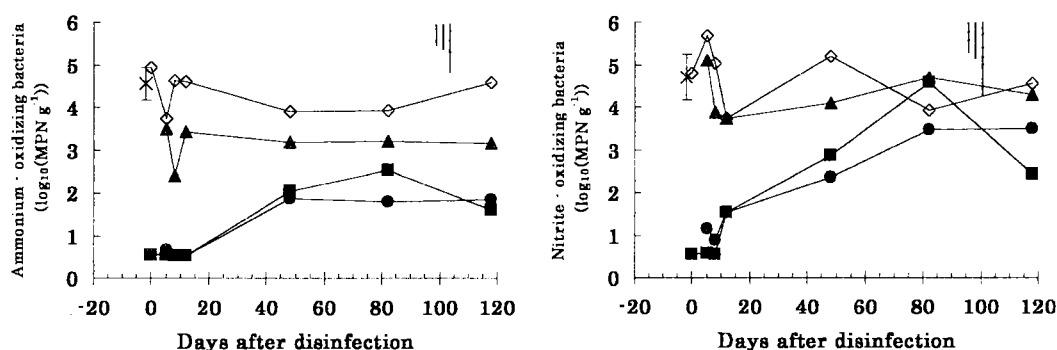


Fig. 3. Levels of ammonium-oxidizing bacteria and nitrite-oxidizing bacteria. X, before disinfection; $\diamond$, CT; $\blacksquare$, SS; $\blacktriangle$, MB; $\bullet$, CP. See footnote to Fig. 2.

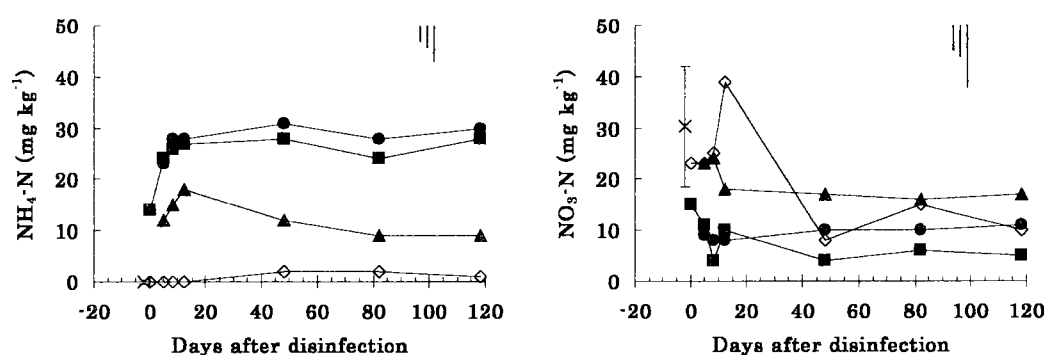


Fig. 4. Levels of $\text{NH}_4\text{-N}$ and $\text{NO}_3\text{-N}$. X, before disinfection; $\diamond$, CT; $\blacksquare$, SS; $\blacktriangle$, MB; $\bullet$, CP. See footnote to Fig. 2.

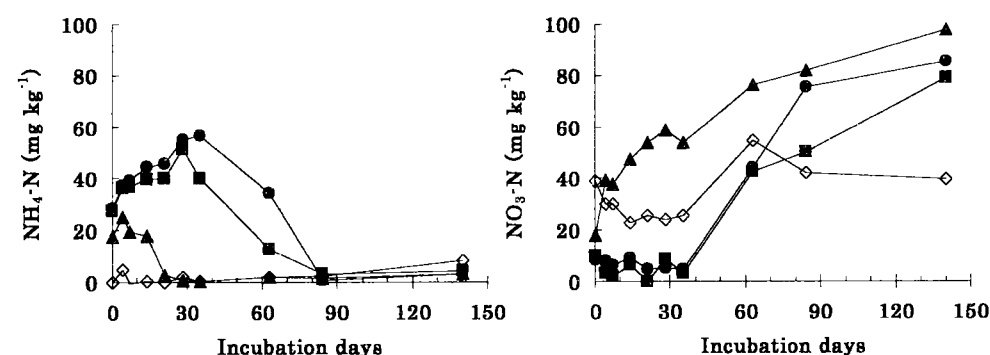


Fig. 5. Nitrogen mineralization patterns of the soils at the time of transplanting (Day 12) determined by the aerobic incubation method. $\diamond$, CT; $\blacksquare$, SS; $\blacktriangle$, MB; $\bullet$, CP. Each plot represents the average values of the results for the treatments with 4 replicates.

nitrification in the MB treatment did not show any time lag and the amount of $\text{NO}_3\text{-N}$ exceeded that in the CT.

Metabolic diversity of microbial communities

The richness in metabolic diversity decreased to a very low value at Day 0 in the SS treatment (Fig. 6), and then rapidly increased, exceeding the values in the other

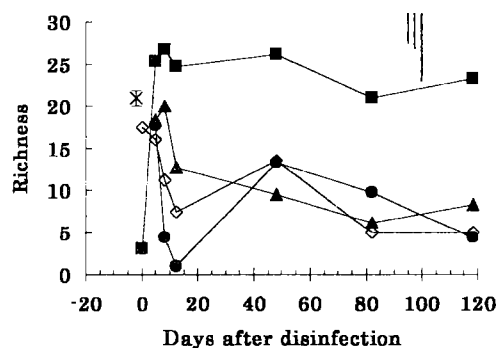


Fig. 6. Richness in microbial communities evaluated by Biolog ECO MicroPlate. X, before disinfection; ◇, CT; ■, SS; ▲, MB; ●, CP. See footnote to Fig. 2.

treatments and it remained higher until Day 118. The value in the CP treatment became minimum at Day 12, while the values in the MB treatment and the CT showed a gradual reduction throughout the experiment. After Day 48, the difference in the richness among the MB and CP treatments and the CT was negligible.

The AWCD value in the CT decreased consistently in the course of the experiment (Fig. 7). Although the tendency of the AWCD value in the MB treatment was similar to that in the CT, the SS and CP treatments gave rather characteristic results. In the SS treatment, the AWCD value was higher after treatment than before treatment at Day -2, except for Day 0 when the value became minimum. In the CP treatment, the AWCD value gradually decreased until Day 12, followed by an increase at Day 48 and then by a decrease at Days 82 and 118.

Plant growth

Tomato plants grew more vigorously in the SS and CP treatments (Table 2). The N content was significantly higher in the SS, CP, and MB treatments than in the CT.

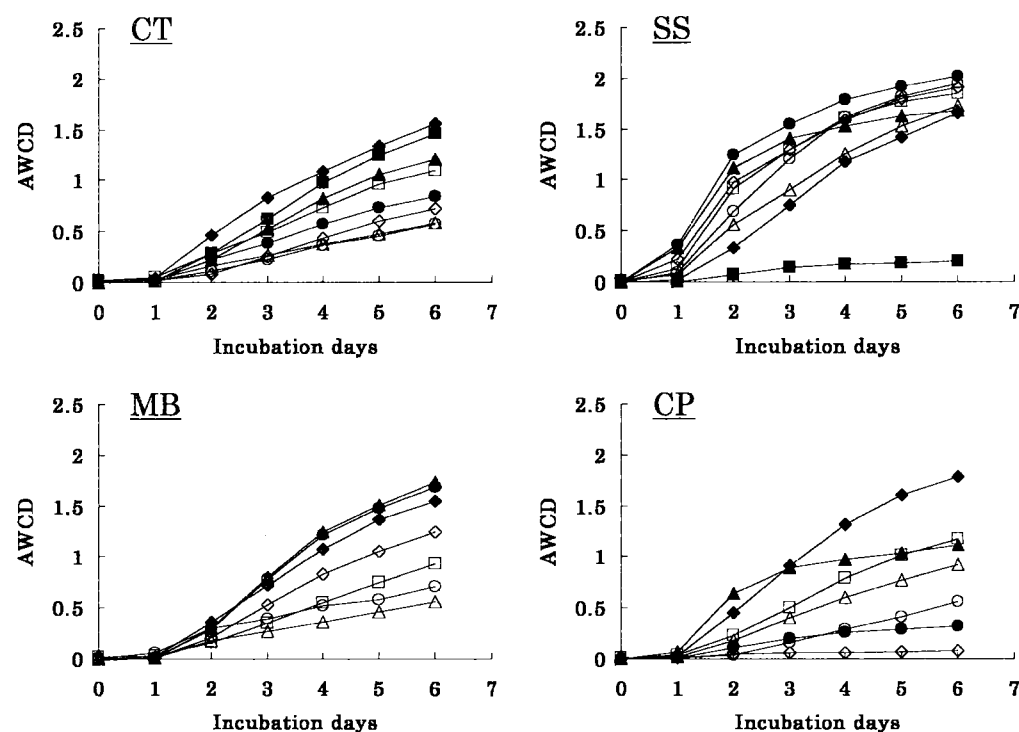


Fig. 7. AWCD value determined by Biolog ECO MicroPlate. Days after disinfection: ◆, -2; ■, 0; ▲, 5; ●, 8; ◇, 12; □, 48; △, 82; ○, 118.

Table 2. Growth of tomato plants 1 month after transplanting.

Treatment	Stem length (cm)	No. of leaves	Leaf area (cm ²)	SPAD	N (g kg ⁻¹)
CT	147 a	21.9 a	553 a	39.2 a	32.4 a
SS	155 b	22.1 a	797 c	41.6 b	38.6 b
MB	148 a	22.0 a	641 ab	39.4 a	36.5 b
CP	154 b	22.3 a	722 bc	39.0 a	37.2 b

Different letters in each column indicate significant difference at the 5% level by the Duncan's multiple range test.

Emergence of weeds during the growth period was negligible in the SS and MB treatments and was very limited in the CP treatment, compared with that in the CT.

DISCUSSION

With the SS treatment, the general physico-chemical properties of the soils did not show any changes or displayed minimal changes in spite of the introduction of a large amount of steam. Based on the fact that the soil water content increased by only 2%, a large proportion of the introduced steam might have cooled down and been transformed into liquid water at a soil depth of more than 0–15 cm where soil sampling was conducted. The increase in the soil pH in the SS treatment could be ascribed mainly to the accumulation of $\text{NH}_4\text{-N}$ and the low activity of nitrifiers (Fujimura et al. 1970).

The viable counts of bacteria remained virtually unchanged by the steam and fumigation treatments, in contrast to the observation by Ridge (1976) using chloropicrin and methyl bromide. However, stimulation of the proliferation of the bacteria after the treatment, compared with the CT, was in agreement with the findings of Ridge (1976) and Toyota et al. (1999; metam sodium). On the other hand, the decrease in the numbers of fungi by the SS, MB, and CP treatments, was in agreement with the report of Shizuoka Natl. Agric. Exp. Stn. (1971; steam sterilization) and that of Itoh (2002; chloropicrin). It is noteworthy that the response of the fungi was different between the SS treatment and the MB and CP treatments. The SS treatment could eradicate fungi only when the soil temperature increased by the steam applicator operation while some part of the fumigants might have persisted in the soils and suppressed the fungi even after the removal of the covering sheet (Siebering and Leistra 1975).

Ammonium-oxidizing bacteria and nitrite-oxidizing bacteria were severely affected by the SS and CP treatments, resulting in their virtual disappearance. The MB treatment also reduced the numbers of ammonium-oxidizing bacteria to a lesser extent than the SS and CP treatments but did not affect those of the nitrite-oxidizing bacteria. The faster recovery of the nitrite-oxidizing bacteria than that of the ammonium-oxidizing bacteria coincided with the results reported by other researchers (Shizuoka Natl. Agric. Exp. Stn. 1971; Toyota et al. 1999). The data on microbial biomass C and N suggested that the SS and CP treatments eradicated microorganisms more efficiently than the MB treatment and that the influence of the former persisted until the end of the experiment. The amount of organic N extracted from non-fumigated soils in the microbial biomass N assay increased by the SS and CP treatments (data not shown).

The increase in the $\text{NH}_4\text{-N}$ plus organic N level between Day 5 and Day -2 was almost equal to and negatively correlated with the decrease in the level of microbial biomass N ($r = 0.754^{**}$, $n = 12$). Therefore, the increase in the $\text{NH}_4\text{-N}$ and organic N levels observed after the treatments could be attributed to the release of the compounds by partial decomposition of the sterilized microbial cells (Brookes et al. 1985). While the level of $\text{NH}_4\text{-N}$ remained high throughout the experiment in the SS and CP treatments, the level of $\text{NO}_3\text{-N}$ was low but remained almost constant along with plant growth in spite of the uptake by plants and leaching. This fact and the increase in the numbers of nitrifiers observed from Day 48 suggested that the activity of the nitrifiers had slightly recovered. In contrast, the MB treatment affected the soil nitrifiers to a lesser extent. Both ammonification and nitrification patterns supported such an assumption on the N dynamics and nitrifier activity in situ. The decrease in the level of $\text{NO}_3\text{-N}$ just after the SS treatment was in agreement with the results obtained by Sonneveld (1979), who assumed that the decrease in the level of $\text{NO}_3\text{-N}$ could be partly ascribed to the reduction to nitrite and the transformation to a gaseous form of N as well as leaching. In this experiment, the loss by leaching appeared to be minimal, based on the relatively small difference in the moisture content of the soils before and after the SS treatment. The mechanism(s) should be further elucidated.

The evaluation of metabolic diversity by the Biolog plate method reflected well the effect of the treatments on the soil microorganisms. The evaluation using the richness index indicated that the metabolic diversity of the soil microbial communities was enhanced by the SS treatment and was not seriously affected by the MB and CP treatments except for the period between Day 8 to 12 in the CP treatment. The other indicators such as substrate diversity and evenness, which were used by Zak et al. (1994), were determined for the data in this study and provided almost the same results as those of richness (data not shown). On the other hand, based on the AWCD values, the effect of the SS, MB, and CP treatments on the soil microbial communities can be explained as follows. In the SS treatment, in spite of the reduced activity of the microorganisms at Day 0, the activity of the surviving microorganisms, including their persistent forms might have been enhanced by the favorable soil environmental conditions, with a low population density and substrate addition from the decomposed microorganisms and soil organic matter through steam heating and high pressure. The CP treatment gradually affected the soil microbial communities from the time of its application to even after the removal of the covering sheet. As indicated by the results on N mineralization and microbial biomass N in this study and the previous

study (Tanaka et al. 2001), the MB treatment was less effective on the eradication of the soil microorganisms. The difference in the effectiveness between the MB and CP treatments observed in this study could be partially ascribed to the fact that chloropicrin has a much lower vapor pressure and a higher water / air ratio than methyl bromide, resulting in a much slower dispersion and evaporation of chloropicrin (Vanachter 1975) and partially to the fact that their application was conducted at the end of summer when the air temperature in the greenhouse was high. Principal component analysis using the absorbance of all the substrates in the Biolog plate after 3 d incubation clearly depicted the rather strong effect of the SS treatment on substrate utilization by the soil microbial communities (Fig. 8). The influence persisted until the end of the experiments. However, the effects of the MB and CP treatments were not clear except for those at Day 5. In spite of the limitation of the application of the degree of metabolic diversity evaluated by the Biolog plate method to microbial functions in situ (Tate 2000), the determination in this study enabled to differentiate the impact of steam sterilization and fumigation on soil microbial communities.

The vigorous growth of the tomato plants in the SS and CP treatments could be attributed to the increase in the N supply at the initial stage although the form of N was ammonium. The physiological disorders due to $\text{NH}_4\text{-N}$ accumulation after the SS, as well as Mn, have

been reported for various plants (Sonneveld 1979), although the adverse effect on plant growth was not obviously observed in our study. For the application of the steam sterilization method in a greenhouse, measures to reduce the accumulation of $\text{NH}_4\text{-N}$ should be developed. The application of organic materials after steam sterilization to promote the recovery of soil nitrifiers, as reported by Morikuni et al. (1999) for tomato cultivation in the isolated beds could be considered. The effect of the application on the microbial communities as well as N dynamics and numbers of nitrifiers should also be investigated because various microorganisms are likely to be introduced to the fields with a lower population density of soil microorganisms after steam sterilization.

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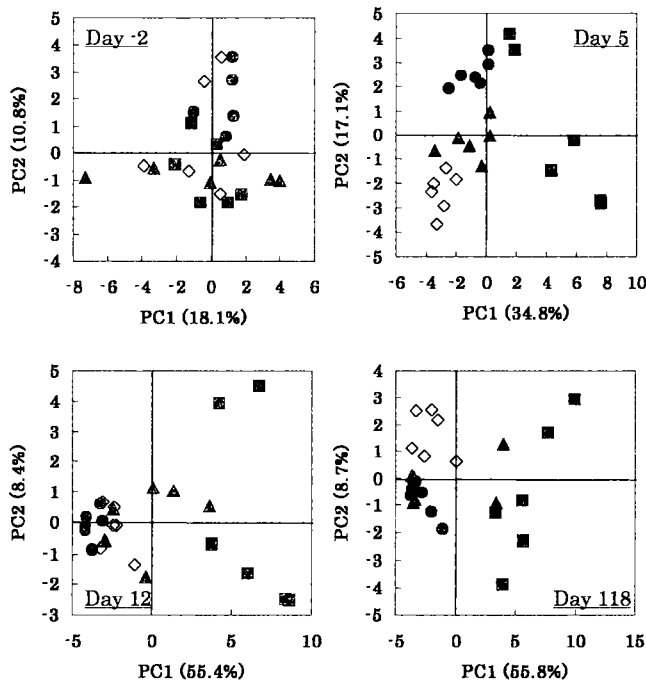


Fig. 8. Principal component analysis for the metabolic diversity of soil microbial communities. ◇, CT; ■, SS; ▲, MB; ●, CP. The percentage in parenthesis represents the contribution for the principal component.

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